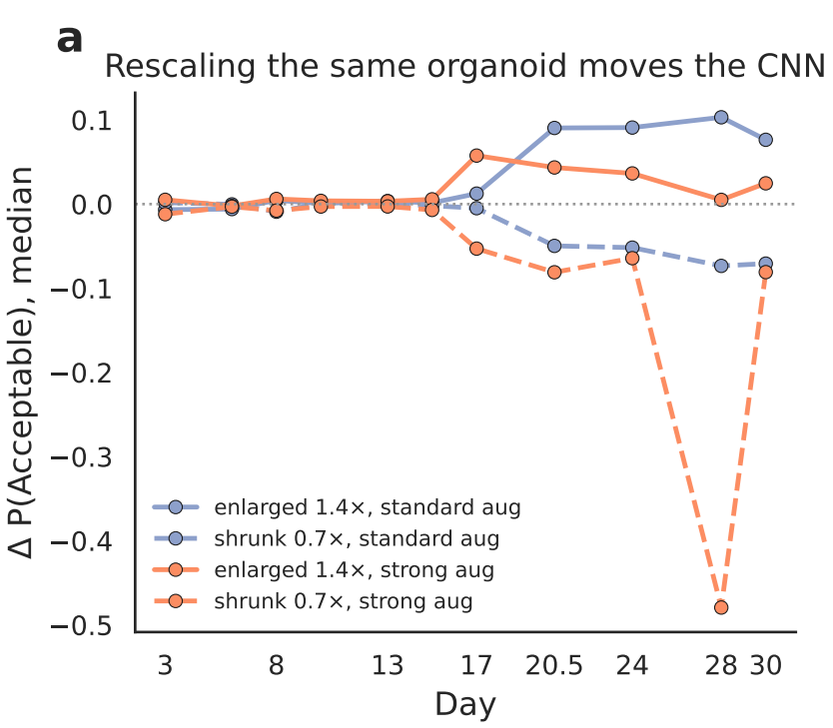
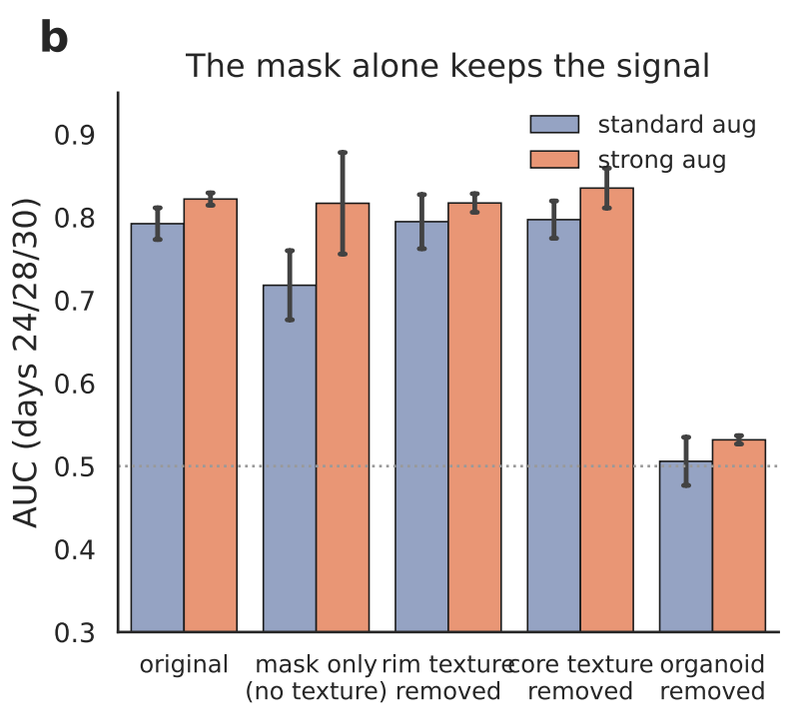
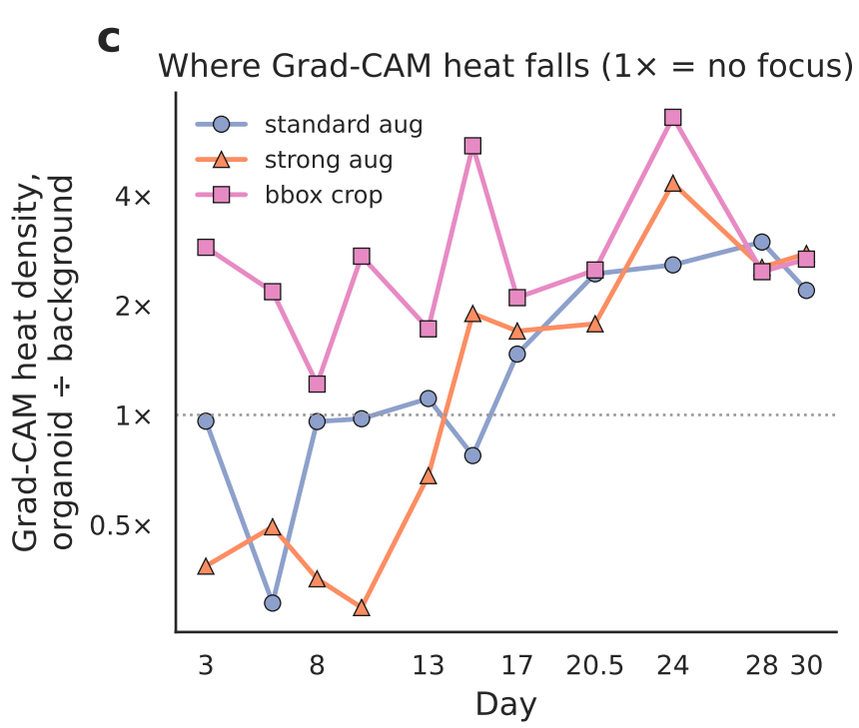
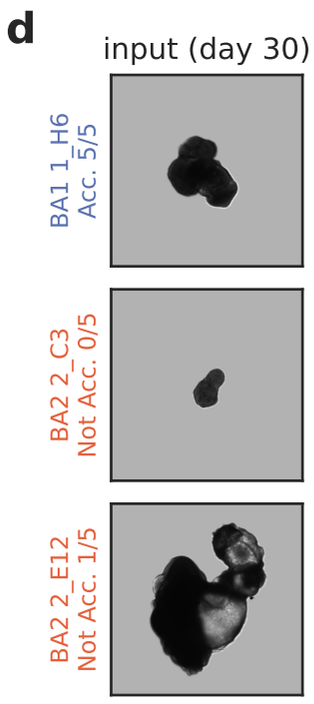
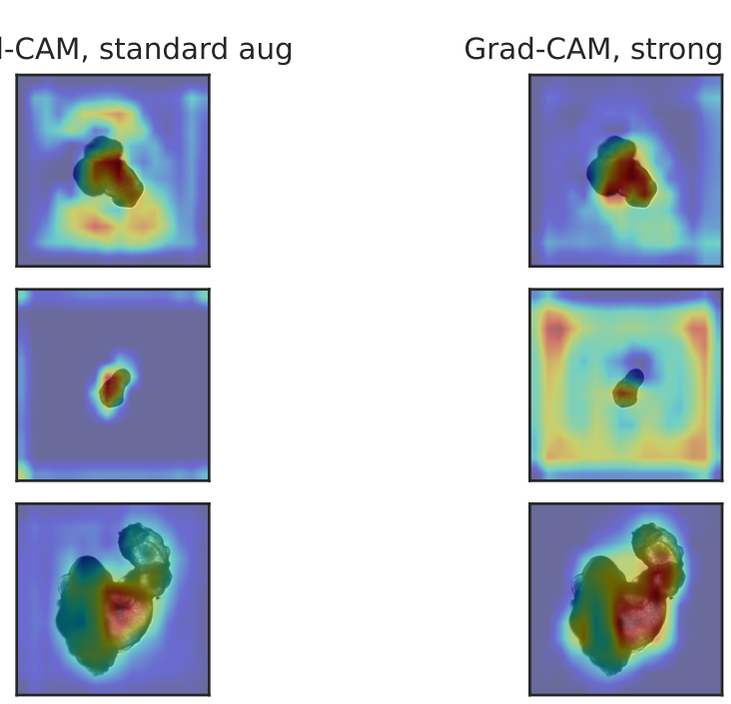

a
Rescaling the same organoid moves the CNN
Δ P(Acceptable), median
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
enlarged 1.4×, standard aug
shrunk 0.7×, standard aug
enlarged 1.4×, strong aug
shrunk 0.7×, strong aug
3
8
13
17
20.5
24
28
30
Day

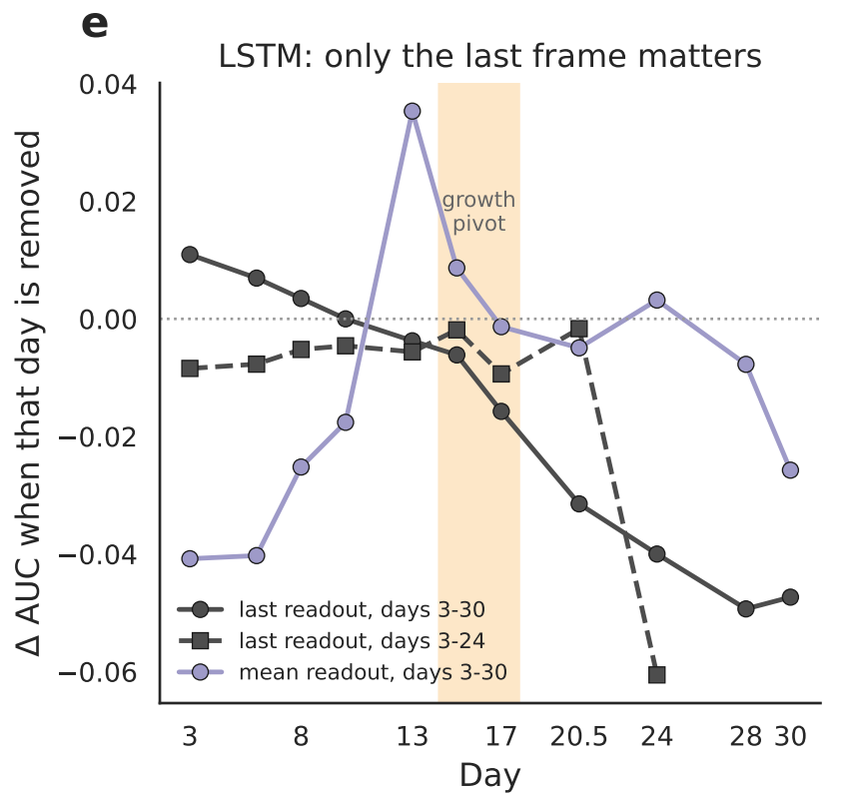

b
The mask alone keeps the signal
AUC (days 24/28/30)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
standard aug
strong aug
original
mask only (no texture)
rim texture removed
core texture removed
organoid removed

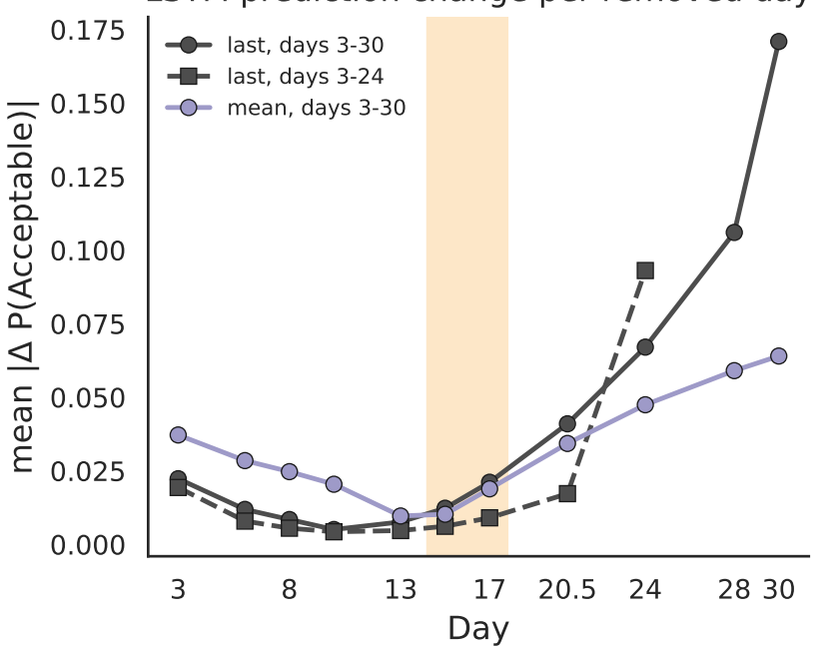

c
Where Grad-CAM heat falls (1× = no focus)
Grad-CAM heat density, organoid ÷ background
4×
2×
1×
0.5×
standard aug
strong aug
bbox crop
3
8
13
17
20.5
24
28
30
Day

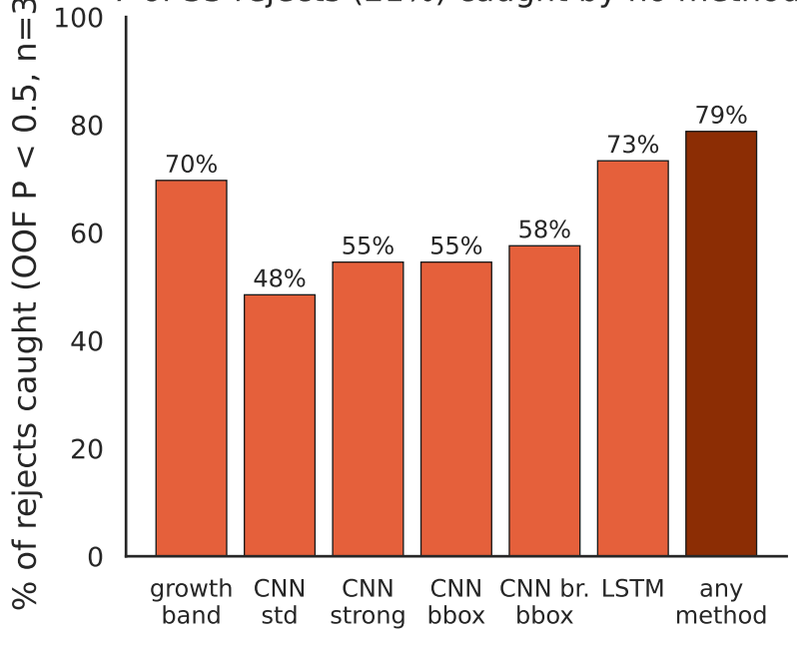

d
input (day 30)
BA1 1_H6
Acc. 5/5
BA2 2_C3
Not Acc. 0/5
BA2 2_E12
Not Acc. 1/5

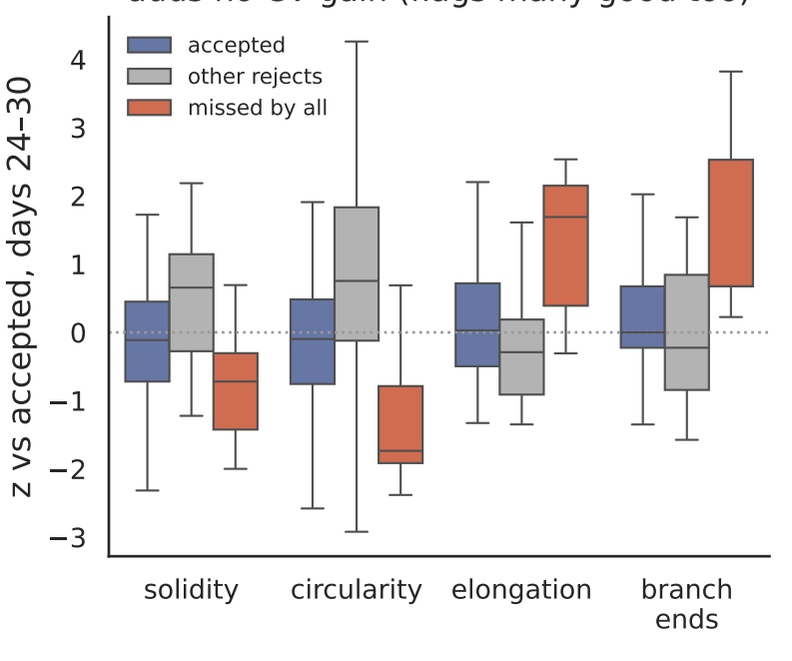

Grad-CAM, standard aug

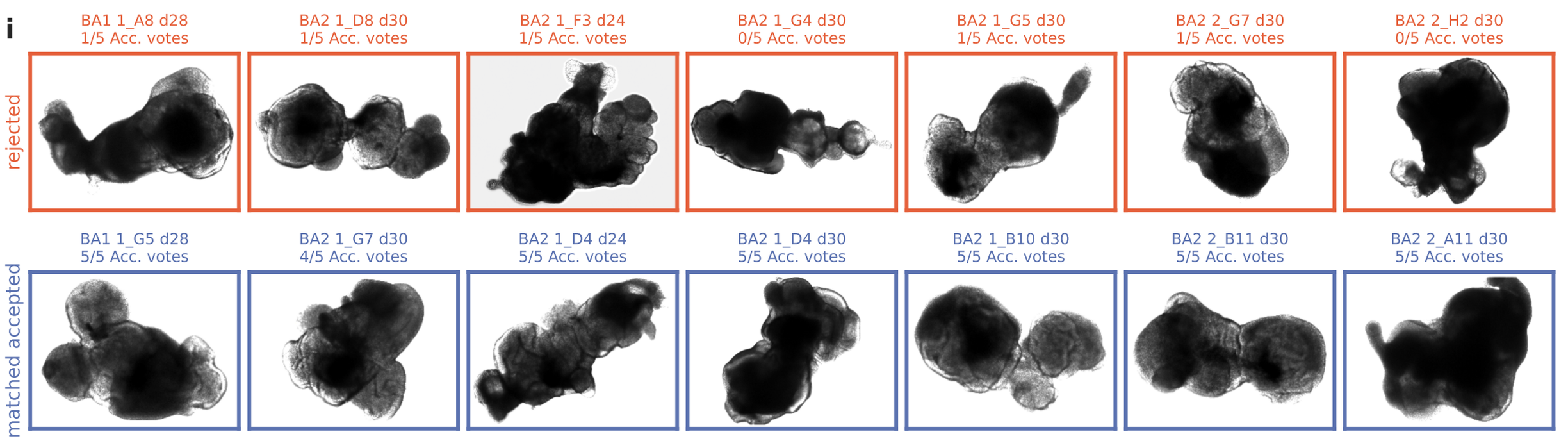
Grad-CAM, strong aug


e
LSTM: only the last frame matters
Δ AUC when that day is removed
0.04
0.02
0.00
−0.02
−0.04
−0.06
growth pivot
last readout, days 3-30
last readout, days 3-24
mean readout, days 3-30
3
8
13
17
20.5
24
28
30
Day


f
LSTM prediction change per removed day
mean |Δ P(Acceptable)|
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
last, days 3-30
last, days 3-24
mean, days 3-30
3
8
13
17
20.5
24
28
30
Day


g
7 of 33 rejects (21%) caught by no method
% of rejects caught (OOF P < 0.5, n=33)
100
80
60
40
20
0
70%
48%
55%
55%
58%
73%
79%
growth band
CNN std
CNN strong
CNN bbox
CNN br. bbox
LSTM
any method


h
Missed rejects are less round, but shape adds no CV gain (flags many good too)
z vs accepted, days 24–30
4
3
2
1
0
−1
−2
−3
accepted
other rejects
missed by all
solidity
circularity
elongation
branch ends


i
rejected
BA1 1_A8 d28
1/5 Acc. votes
BA2 1_D8 d30
1/5 Acc. votes
BA2 1_F3 d24
1/5 Acc. votes
BA2 1_G4 d30
0/5 Acc. votes
BA2 1_G5 d30
1/5 Acc. votes
BA2 2_G7 d30
1/5 Acc. votes
BA2 2_H2 d30
0/5 Acc. votes
matched accepted
BA1 1_G5 d28
5/5 Acc. votes
BA2 1_G7 d30
4/5 Acc. votes
BA2 1_D4 d24
5/5 Acc. votes
BA2 1_D4 d30
5/5 Acc. votes
BA2 1_B10 d30
5/5 Acc. votes
BA2 2_B11 d30
5/5 Acc. votes
BA2 2_A11 d30
5/5 Acc. votes